

Quasi-frozen spin for deuteron and proton beam at EDM storage ring

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A novel experimental setup with spatially separated magnetic and electrostatic fields is proposed for studying the electric dipole moment (EDM) of deuterons and protons. The design leverages Wien filters to achieve precise measurements under "quasi-frozen" spin conditions, demonstrating adaptability to existing facilities for research at higher energy levels.

Keywords: Wien filter, Electrostatic deflector, Quasi-frozen spin, Electric dipole moment

I. INTRODUCTION

The investigation of the electric dipole moment (EDM) of fundamental particles, such as protons and deuterons, constitutes a critical endeavor in contemporary physics. EDM offers an unparalleled opportunity to probe CP violation and validate theoretical models that describe particle interactions beyond the framework of the Standard Model. For charged particles, the precise measurement of EDM necessitates the use of accelerators and the development of novel methodologies and technologies, requiring the adaptation of existing accelerator facilities to meet these experimental demands.

This paper explores a versatile structure characterized by spatially segregated magnetic and electrostatic fields, uniquely designed to facilitate the study of EDM in both deuterons and protons. A notable feature of this structure is its ability to support experiments for both types of particles, thereby enhancing its applicability and efficiency. Special emphasis is placed on the utilization of Wien filters to compensate for the magnetic dipole moment (MDM) and establish conditions conducive to a "quasi-frozen" spin. The analysis elucidates how such a structure can be adapted for facilities not originally intended for such experiments and explores its potential for advancing research at higher energy levels.

$$\begin{aligned} \frac{d\vec{S}}{dt} &= (\vec{\Omega}_{\text{MDM}} + \vec{\Omega}_{\text{EDM}}) \times \vec{S}, \\ \vec{\Omega}_{\text{MDM}} &= -\frac{q}{m\gamma} \left\{ (\gamma G + 1) \vec{B}_{\perp} + (G + 1) \vec{B}_{\parallel} - \left(\gamma G + \frac{\gamma}{\gamma + 1} \right) \frac{\vec{\beta} \times \vec{E}}{c} \right\}, \\ \vec{\Omega}_{\text{EDM}} &= -\frac{q\eta}{2m} \left(\vec{\beta} \times \vec{B} + \frac{\vec{E}}{c} \right), \quad G = \frac{g - 2}{2}, \end{aligned} \quad (1)$$

where $\vec{\Omega}_{\text{MDM}}$, $\vec{\Omega}_{\text{EDM}}$ – are the angular speed of rotation due to magnetic dipole and electric dipole moment, E , B – external magnetic and electric field.

For EDM measurements, first necessary to compensate MDM-effect. The simplest idea to fully compensate magnetic with electric field along the momentum. In such way, the lattice should be exclusively build for EDM research.

Alternative, quasi-frozen spin [1] based on the principle of sequential spin compensation. This way let In this case EDM measurement $S_{\text{EDM}} = 1 - \frac{\alpha^2}{4} \simeq 0.998$ [1]. Here α is small angle of MDM effect.

A. Relative spin motion at QFS

For QFS we should first maintain close orbit condition

$$\Phi_p^{\text{arc}} + \Phi_p^{\text{comp}} = \frac{2\pi}{N} \quad (2)$$

where index p states for momentum, and Φ_p^{arc} and Φ_p^{comp} – total momentum rotation in arc and spin compensator element, N is the lattice periodicity.

The second condition is to compensate MDM spin rotation in ring

$$\Phi_s^{\text{arc}} + \Phi_s^{\text{comp}} = 0 \quad (3)$$

where Φ_s^{arc} and Φ_s^{comp} – total spin rotation in arc and spin compensator.

II. QUASI-FROZEN SPIN AT PERIODIC LATTICE

For an ansamble of particles, spin describes as a vector and evaluate in external EM-fields as a classical value with T-BMT equation [1]

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If we consider a ring with $\Phi_p^{\text{arc}} = \frac{2\pi}{N}$, then $\Phi_p^{\text{comp}} = 0$, so it should not violate orbit at all. From the eq. 2 for spin compensation element with both magnetic and electric

$$\Phi_{pB}^{\text{comp}} + \Phi_{pE}^{\text{comp}} = 0 \quad (4)$$

where Φ_{pB}^{comp} and Φ_{pE}^{comp} – total momentum rotation in magnetic and electric field at spin compensator.

The ratio between orbital and spin rotation as follows given by spin-tune in magnetic field $\Phi_s^{\text{arc}} = \nu_B \Phi_p^{\text{arc}}$, $\Phi_{sB}^{\text{comp}} = \nu_B \Phi_{pB}^{\text{comp}}$ and electrical $\Phi_{sE}^{\text{comp}} = \nu_E \Phi_{pE}^{\text{comp}}$

Finally, for rotation via magnetic field in spin compensator element

$$\Phi_{pB}^{\text{comp}} = -\Phi_{pE}^{\text{comp}} = -\Phi_p^{\text{arc}} \frac{2\pi}{N} \frac{\gamma^2 G}{G+1} \quad (5)$$

No mention was made of the physical structure of the compensation element, only of the integral features of the presented field components. So, the principles described here can be applied to both a Wien filters and an electrostatic deflectors with magnetic kickers.

B. Spin compensation element length

Spin compensation element length can be calculated for magnetic and electric field

$$L = \Phi_{pB}^{\text{comp}} R_B = \Phi_{pE}^{\text{comp}} R_E \quad (6)$$

where R_B, R_E – radius of magnetic and electric field. In general, total radius for element with combined magnetic and electric fields can be find as

$$\frac{1}{R} = \frac{1}{R_B} + \frac{1}{R_E} \quad (7)$$

where $B\rho = \frac{p_0}{e}$ – magnetic rigidity, $p_0 = \gamma m \beta c$, $\kappa = \frac{p_0 \beta c}{e}$ – electrical rigidity. If we consider no orbital distortion in total, Lorentz force should be zero, so $R = 0$ and $R_B = -R_E$.

The limitation on the strength of the electric field is much more significant than in the case of a magnetic field. According to the eqs. 5 and 6

$$L_{\min} = \Phi_{pE}^{\text{comp}} R_E = -\frac{2\pi}{N} \frac{\gamma^2 G}{G+1} \frac{\kappa}{E_{\max}} \quad (8)$$

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III. FEATURES OF PROTON AND DEUTERON

A. Dependence on particle G

can be analyzed only at polarimeter.

Orbital and spin dynamic differs for various sort of particles. As mass-to-charge ratio twice time bigger for deuterons, then protons

Anomalous magnetic moment for proton $G_p = 1.79$ and deuteron $G_d = -0.14$

B. Wien filter

Wien filter consist of crossed fields and make no deflection along beamline. This make difficulties, as accurate additional space at the ring. Moreover, as we can see at eq. 5

And the same structure can be used to study protons, but with Wine filters rotated 180 degrees at a lower energy.

C. Electrostatic deflector with kicker

Using the electrostatic deflector necessary to accurate kicker usage. So the length become longer for an additional kicker. Such element can be used for creation of bypass section in addition to native straight section.

IV. SUMMARY

In this article shown a principle of quasi-frozen spin lattice, main adventure of such structure can be investigation both proton and deuteron beam. As it is QFS, lattice construction also allow to use ring for research at higher energy levels.